Latent Multi-Criteria Ratings for Recommendations

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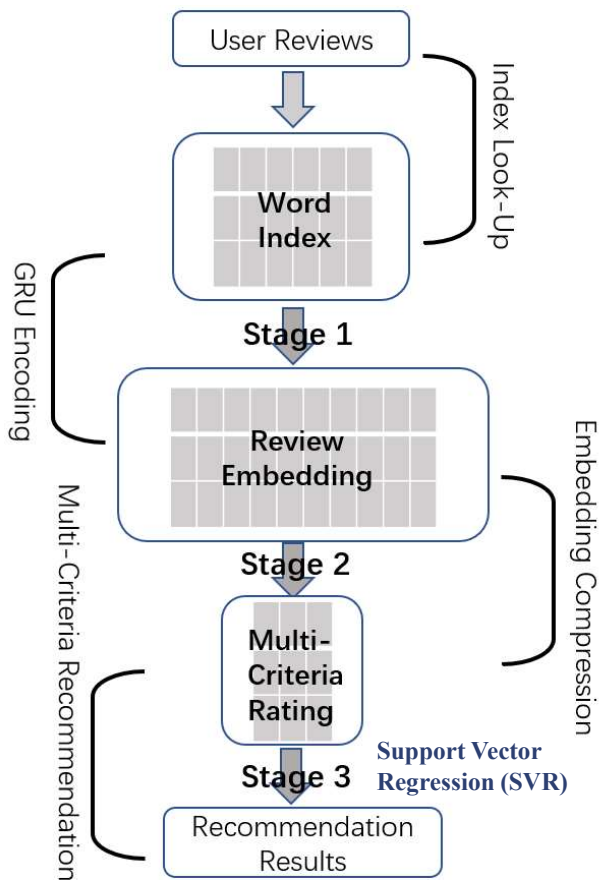
Problems of Explicit Multi-Criteria Ratings

- **Suffer from the missing value problem**
- **Can be noisy and sparse**
- **Might not be sufficient to capture complex user preferences**

Solution: LatentMC

- **Utilize user reviews to generate latent multi-criteria ratings**
- **Compress high-dimensional review embeddings to low-dimensional discrete ratings using Gumbel-Softmax reparameterization**

Model



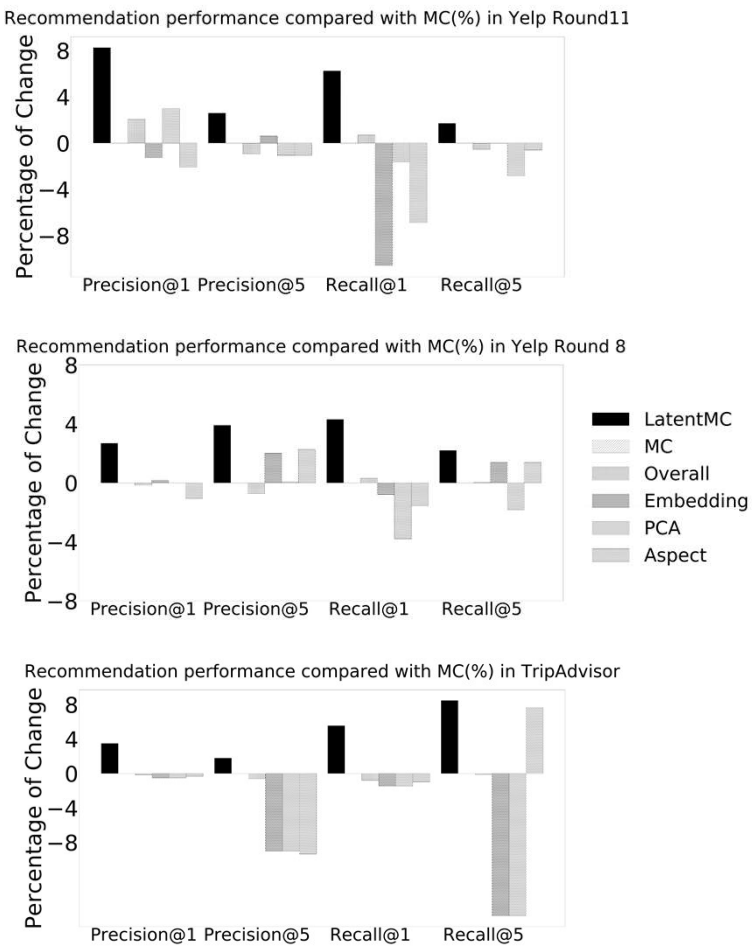
Experiments

Datasets

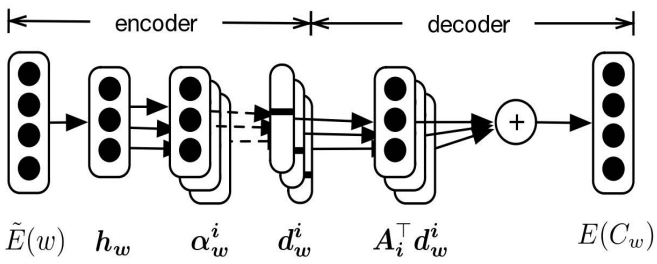
- Yelp Challenge Round 8
- Yelp Challenge Round 11
- TripAdvisor Dataset

Baselines (compared with our LatentMC model)

- **MC:** explicit multi-criteria ratings
- **Overall:** overall ratings only
- **Embedding:** uncompressed review embeddings
- **Aspect:** extracted aspect ratings [BLT17]
- **PCA:** top-k factors extracted using Principal Component Analysis



From review embeddings to latent multi-criteria ratings



Conclusions

- **Latent multi-criteria ratings perform *consistently* and *significantly* better than other baseline models.**
- **Compression and reparameterization play crucial roles in improving multi-criteria recommendation accuracy. (comparing LatentMC with the Embedding baseline)**